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Utilizing First and Second Derivative Seismic Trace Amplitudes to Map Paleo-Glacial Valleys

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Abstract

Explaining variation in reservoir thickness from well data and geological models can be challenging when dealing with non-uniform depositional nature. A common example is glacial valley distribution cutting into deep marine shale. Reservoir thickness varies tremendously at the same depth in valley centers, valley edges or outside the valley. Moreover, valleys have different geometries including sharp edges (U-shape) valleys and softer edges V-shape valleys. The spatial variation of geometrical change also introduces another aspect of complexity in mapping these valleys. The architecture of glacial valleys cutting into shaly stratigraphic units has the advantage of soft to stiff contrast which can be imaged by 3-D seismic. The glacial process of carving melted ice and sediments into shale is chaotic, leaving an imprint of elongated valleys that can be mapped in seismic stratigraphic slices and seismic cross sections. On seismic 2-D cross sections, the valley walls can be traced with limitations due to the seismic quality and complexity of structuration. Advanced seismic interpretation methods are utilized to enhance the cross-section resolution and reduce the uncertainty in mapping the glacial valleys. These methods include deriving seismic trace first and second derivatives to sharpen the amplitude contrast in viewing the seismic images. Advancing the glacial valleys mapping techniques has a direct impact on updating gross depositional environment maps and interpreting the paleo topography.

Geological background

The Arabian Plate has encountered two major glaciation events. The late Ordovician to early Silurian glacial retreat and advance, and the late Carboniferous to early Permian post-Hercynian glacial period. Outcrop evidence of glaciation is recorded in multiple areas such as Rahal Dhab (Ordovician) and Wajid region (Carboniferous), [Michael et al 2018](#) and [Melvin and Norton 2013](#), (figure 1, A &B). In addition, subsurface images and cores show tunnel valley structuration and diamictite facies of glaciomarine depositional environment (figure 1, C, [Melvin and Norton, 2013](#)).

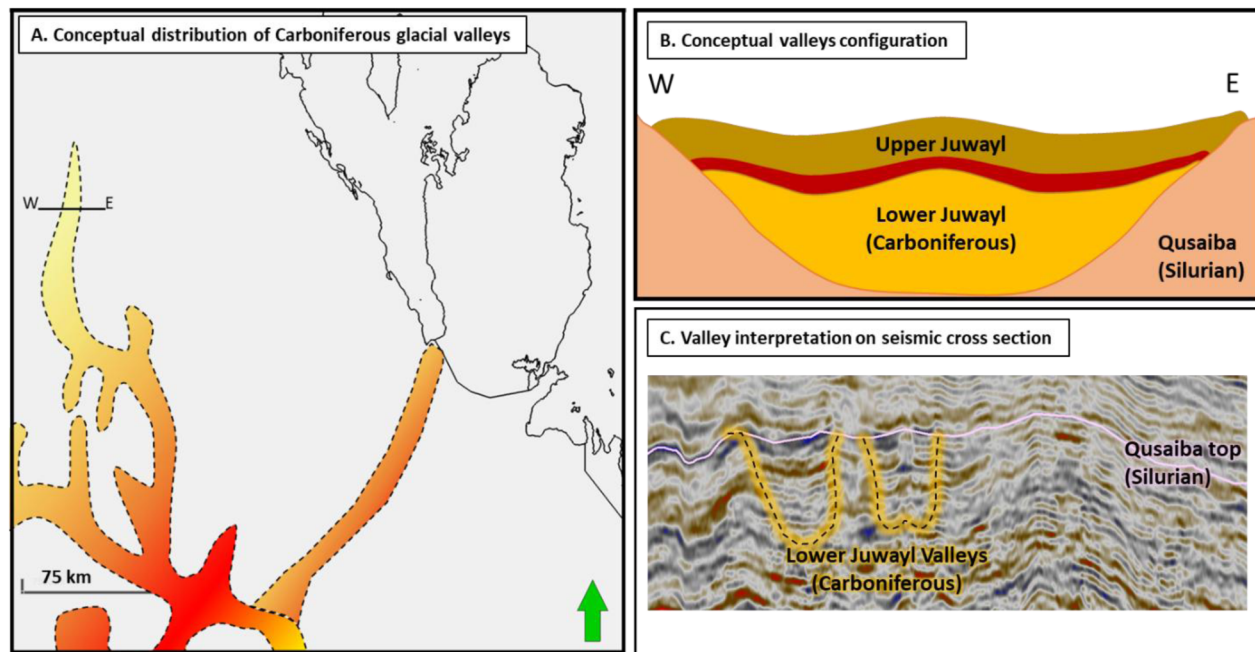


Figure 1—a) Base map with conceptual distribution of Carboniferous age glacial valleys based on [Melvin and Norton's interpretation, 2013](#). b) Schematic cross section with conceptual Carboniferous valley configuration cutting through the Silurian age Qusaiba Member. The high vertical column of accommodation space is filled with syn-glacial deposits of lower Juwayl Formation followed by fluvial glacial deposits of upper Juwayl Formation. c) Seismic cross section with east-west orientation showing the Qusaiba Member top carved by lower Juwayl Formation valleys with distinctive discontinuity in the reflectivity and disturbed signal of the valley fill.

During the Carboniferous period, 320-300 million years ago, the Arabian Peninsula, including Saudi Arabia, was part of the supercontinent of Gondwana ([Keller et al, 2011](#)). Glacial processes during this time are believed to have been influenced by the formation of large ice sheets on Gondwana. The late Carboniferous Juwayl Formation in Saudi Arabia is thought to have been deposited in a glaciogenic environment, with sediments formed as a result of glacial erosion and deposition. The glacial process likely involved the advance and retreat of ice sheets, which carved out valleys and created glacial lakes. The resulting sediments were deposited in a variety of environments including glacial, fluvial, and lacustrine settings. Although data from outcrops and subsurface support the presence of these glacial events, there are inherited seismic resolution challenges when it comes to mapping the glacial paleo-valleys from 3D seismic data.

The tectonic setting of the region played a role in controlling the orientation and shape of the Carboniferous aged valleys. The Arabian Plate was affected by multiple tectonic events during the Carboniferous period, including rifting and faulting, which influenced the direction of fluvial flow, ice advance and retreat and the resulting valley shape. The lithology of the underlying rocks controlled the shape and fill of the Carboniferous age valleys. The rocks of the Carboniferous in Saudi Arabia include sandstones, shales, and limestones, which have different resistances to erosion. The softer rocks, such as shales, were more easily eroded, while the harder rocks such as sandstones were more resistant. The fill of the Carboniferous age valleys is composed of glacio- fluvial sediments, including sandstones, shales, and conglomerates. The type and distribution of these sediments were controlled by the syn-glacial and post-glacial fluvial environment and the underlying topography ([Khalifa, 2015](#)).

Methods

High-resolution 3-D seismic data provide detailed information on the subsurface structure and stratigraphy. The seismic is processed and interpreted to identify tunnel valley features, including geometry, orientation, and infill characteristics ([Kirkham et al, 2021](#)). Seismic stratigraphic analysis is done to reconstruct the

geological history of the tunnel valley, including the timing and mechanisms of formation. This involved correlating the seismic data with other geological and geophysical data such as well logs, cores, and outcrop observations. Characterizing the infill of the tunnel valley, including the type of sediments, thickness, and distribution is inferred from the seismic attributes.

Preparing the seismic data for analysis, included data conditioning, noise reduction, and data formatting. Enhanced removal of multiples can over-smooth the data and remove meaningful detail from the seismic volume. Therefore, the volume used in this study is conditioned without de-multiple techniques. Coherence attributes from the seismic data, which involve measuring the similarity or continuity of seismic reflections between adjacent seismic traces is used for fault and edge mapping (Chopra and Marfurt, 2018). This can be done using various algorithms, such as cross-correlation and semblance. The first derivative volume represents the rate of change of coherence with respect to the spatial coordinates.

A gradient operator is used to compute the first derivative of seismic attributes. The gradient operator is applied in the x, y, and z directions to compute the gradient components (Chopra and Marfurt, 2013). Gradient operators, such as the Sobel or Prewitt operators are designed to estimate the gradient of an image or seismic attribute and are commonly used in image processing and computer vision applications. Gradient operators are more robust to noise than finite differences and can provide more accurate results, especially for complex geometries.

The finite difference method is used to compute the second derivative (Laplacian) of seismic attributes. The Laplacian is computed by applying the finite difference method twice to the seismic data. This method involves approximating the derivative using differences between adjacent samples in the seismic data. The finite difference method is simple to implement and computationally efficient, but it can be sensitive to noise.

The mapping workflow included three phases dependent on data availability: 1) seismic time slices interpretation. 2) seismic cross section interpretation and 3) incorporating non-seismic methods if available (figure 2). A common technique for mapping glacial paleo-valleys is by flattening the seismic cube on the horizon equivalent to or near the glacial event. The aim of flattening is to create a restoration which resembles the architecture at the time of deposition, providing a basis to interpret the depositional environment. To enhance the image of the flattened cube first and second derivatives of the seismic traces were produced.

Four wells were used to calibrate the mapping of the valleys on a post stack time migrated volume slices and cross sections. Wells A, B and D encountered no reservoir (zero thickness), meanwhile, well C encountered a thick section of the reservoir (200 feet). The seismic volume is flattened on the mapped Lower Pennsylvanian reflector (Upper Carboniferous) to restore the structuration. Post flattening, first gradient derivative and second finite derivative were computed from the flattened volume. Slices and cross sections of the different volumes were interpreted to refine the valley outline mapping.

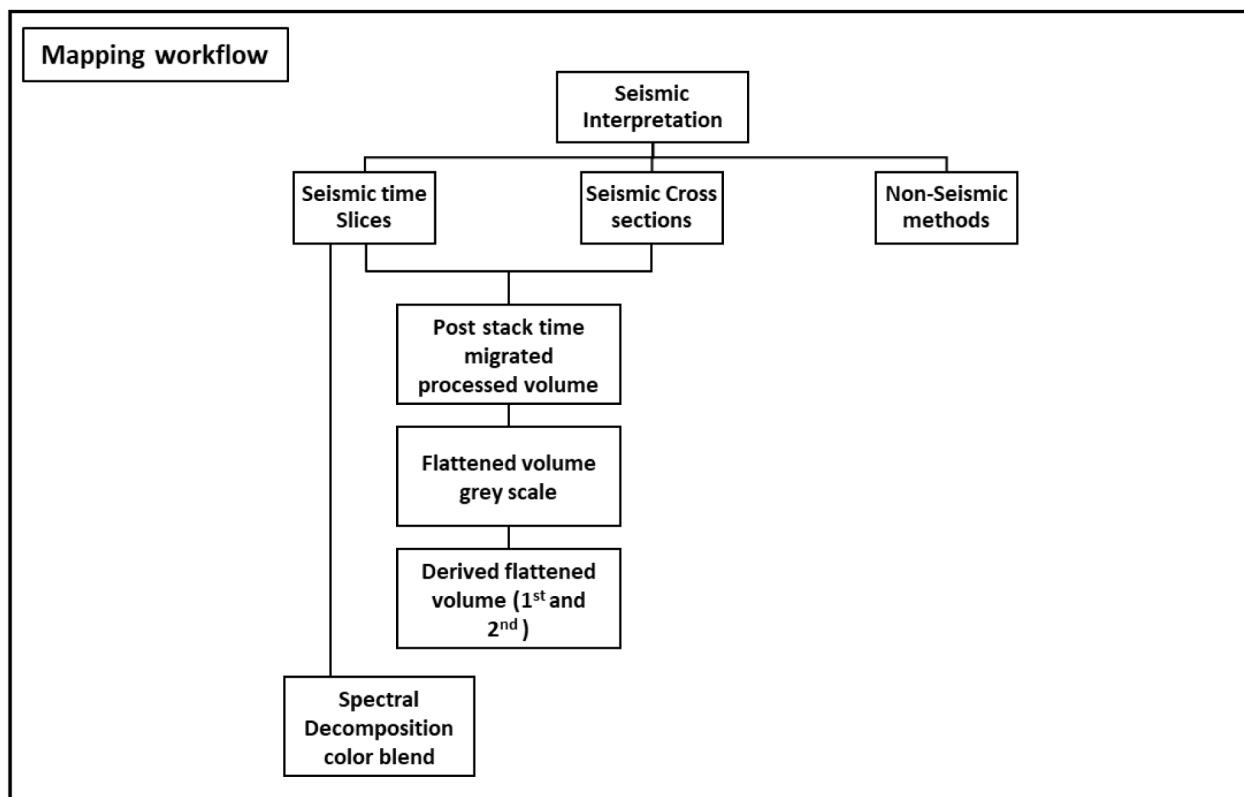


Figure 2—Phases of paleo-glacial valley mapping from subsurface data including seismic time slice interpretation, seismic cross section interpretation and non-seismic methods extractions.

Results

Distinctive features were observed by applying various interpretation workflows. The spatial distribution of paleo-glacial valleys was tracked and a map of valley versus no valley background produced (figure 3, A). Seismic cross-sections from the derived flattened, amplitude first derivative and amplitude second derivative cubes showed improvements in capturing the valley fill signature (figure 3). The appearance of a traceable pronounced reflector helped identify the top of the paleo-valley through the seismic cube. This top reflector shows the contrast between the localized valley soft fill and the overlaying tight upper seal. Valley fill was imaged in the amplitude first and second derivatives cross sections as uneven reflectivity below the valley top.

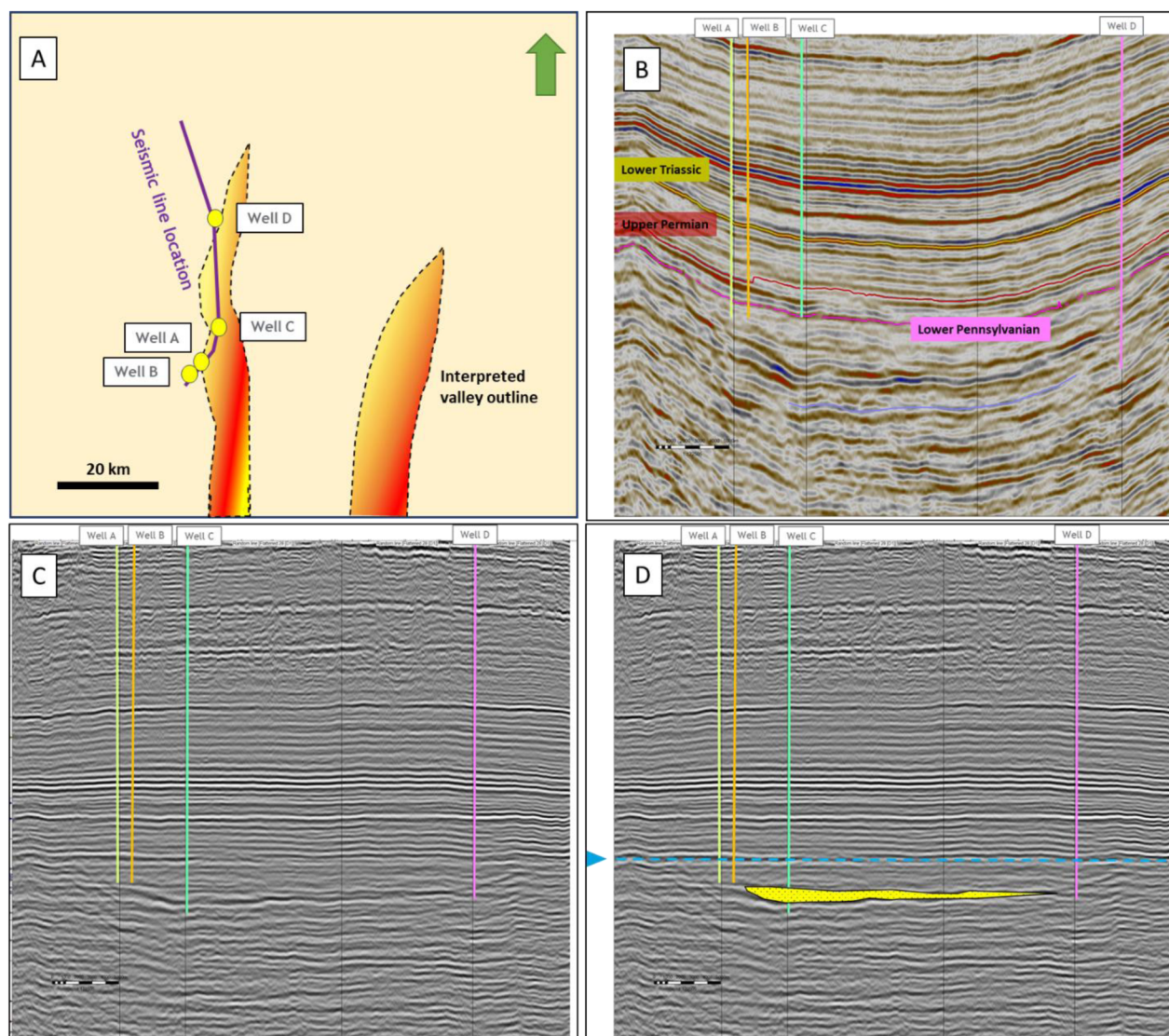


Figure 3—a) Location map of three wells, wells A, B and D confirmed no glacial valley deposits and well C penetrated a thick section of the glacial valley. b) Post stack time migration composite cross section through the three wells. c) A cross section from the flattened first derivative volume showing the signature of the valley cutting into the lower shaly stratigraphy. d) The yellow body represents the glacial valley outline interpreted on cross-section C. The dashed line shows the level on which the section is flattened.

The chaotic signature inside the valley resembles the conglomeratic nature of the destroyed sedimentary layer and the soft sediments transferred by the ice and deposited later in the valley. In addition, noticeable velocity pull-up at the interpreted base of the valley was observed reflecting the contrast between valley fill and surrounding tight rock (figure 4). Stratigraphic slices from the flattened volume, amplitude first and second derivative volumes and spectral decomposition volume were generated to trace the spatial continuation of the glacial valleys. Moreover, a coherence volume of the first and second derivative cubes refined the edges of the paleo-valleys in areas of sharp change in the reflectivity due to the abrupt melting event.

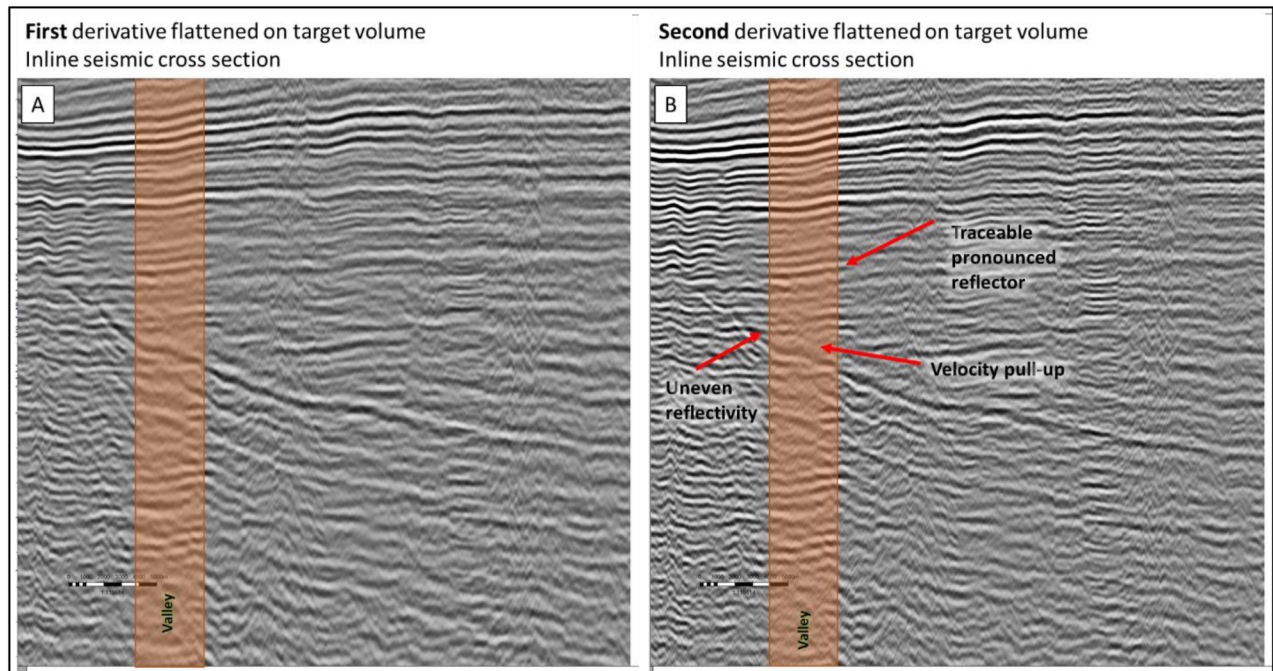


Figure 4—a) first derivative flattened volume on interpreted target top (time equivalent to the top of the valley fill and base of the overlying silty section. b) same line as A with second derivative.

The flattened stratigraphic slice showed lateral change in amplitude (figure 5). The amplitude first derivative minimized the window of the computed traces and preserved the valley outline. Furthermore, the second derivative reduced the amplitude discrepancy outside the valley outline due to calm shale sheet nature and enhanced it within the valley fill to represent the deformation at the valley location. Spectral decomposition slices of mixed frequencies (10, 12, and 14 hertz) was used to calibrate the mapping from the flattened stratigraphic slices. The overlapping frequencies are imaged as bright localities and interpreted as non-valley areas. Dark stripes represent the absence of the common frequencies and interpreted as valley edges.

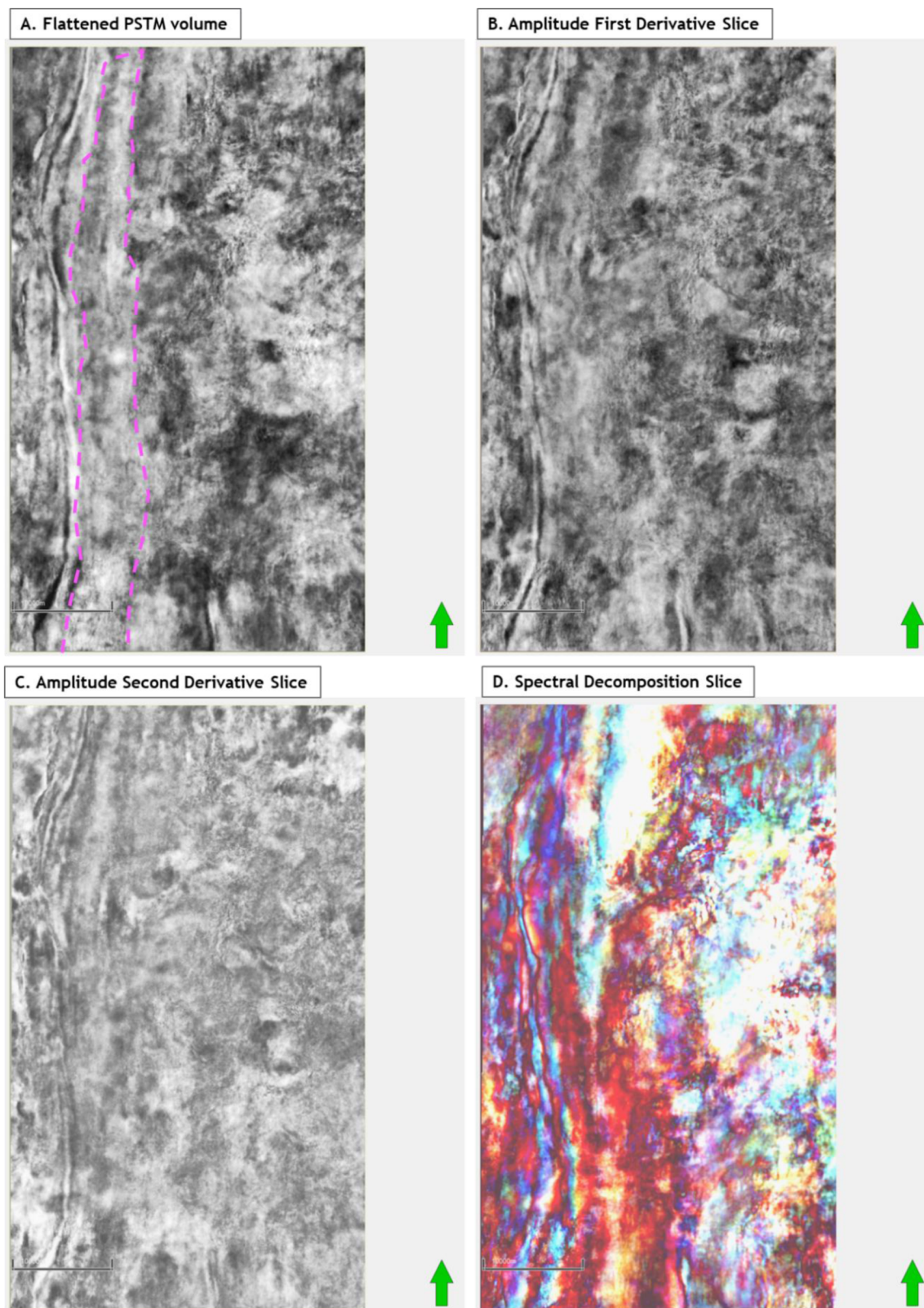


Figure 5—Stratigraphic time slices: a) flattened post stack time migrated slice with valley outline dashed, b) Amplitude first derivative slice, c) Amplitude second derivative slice and d) Spectral decomposition slice.

Discussion

Mapping the extent of glacial valleys is essential for improving gross depositional environment maps, improving understanding of valley direction and to interpret localized highs and the main highland at the glacial advance time. The amplitude first and second derivative volumes are commonly used for fault and structural edge mapping. However, developing these volumes for mapping glacial paleo-valleys is unique and enhances the mapping clarity and confidence compared to the post-stack migration volume before conditioning.

The first derivative of seismic trace amplitude highlights the edges or boundaries of geological features such as faults, fractures, and layer interfaces. This is because the derivative is sensitive to changes in amplitude, which occurs at the edges of these features. By emphasizing these edges, the first derivative can help to improve the visibility of subtle features that may not be apparent in the original seismic data. The second derivative of seismic trace amplitude can further enhance the visibility of subtle features by highlighting the curvature or shape of the seismic reflectors. This can be particularly useful for identifying features such as anticlines, synclines, and other structural features that may not be clearly visible in the original data. The derivatives of seismic trace amplitude can also help to reduce noise in the data. Random noise tends to have a high frequency content, which is suppressed by the derivative operators. As a result, the derivatives can help to enhance the signal-to-noise ratio (SNR) of the data, making it easier to interpret.

By highlighting the edges and curvature of seismic reflectors, the derivatives can help to improve the resolution of the data. Derivatives are sensitive to boundary effects, such as edges or discontinuities in the data. The north-south detected geometry of the valley shows thin a northern tail and thicker southern section (figure 3). The base of the valley is quite distinctive and shown as a high amplitude peak due to the contrast between the soft sediments inside the valley and the hard shale outside of it (Bard and Bouchon, 1980). The base peak has higher amplitude below the thickest interpreted valley fill which is interpreted soft thick section of reservoir. The non-uniform north-south thickness can be a result of heterogeneity in the shale brittleness (Mustafa et al, 2019). The more brittle shale to the north was easier to fracture and make bigger accommodation space for the ice and sand to deposit.

The valley outline produces a reflectivity shadow cutting through shallower and deeper layers compared to the reservoir section. However, valley outline is calibrated by well data as wells A, B and D has no reservoir encountered. The valley has V-shape outline (figure 4), interpreted as glacio fluvial valley. In the presence of valleys only structured by the ice, the valley edges are sharper and detected as U-shape valleys. However, the ice here is flowing in fluvial process making the valley outline gradual in thickness change. V-shaped valleys are primarily formed by fluvial erosion, where rivers cut through the landscape over time (Hirano and Aniya, 1988). The constant flow of water erodes the rock, creating steep sides and a narrow bottom, giving the valley its characteristic "V" shape in cross-section. The river's erosional power is concentrated at the bottom, leading to steeper sides. The type of rock affects the valley's shape; harder rocks result in narrower valleys with steeper sides, while softer rocks lead to wider valleys.

The typical U-shaped valleys are formed by glacial activity. Glaciers move slowly, exerting immense pressure that fractures and plucks rock from the valley walls, resulting in a wider and more rounded shape. The floor is typically flat or gently sloping with steep sides. Glacial plucking and the transport of rock debris contribute to the valley's shape. The duration and extent of glaciation influence the valley's pronounced U-shape unlike what is mapped in the seismic cross sections.

On the seismic cross sections, valleys may appear as depressions or disruptions in the otherwise continuous seismic reflections, indicating a change in subsurface structure. Changes in amplitude can highlight the presence of different sediment types within the valley (Buker et al, 1997). This attribute can reveal disruptions in the subsurface, such as the edges of a valley. The orientation of seismic reflections can indicate the slope of valley sides which can be calibrated by image log interpretation. Sand dip direction is interpreted to estimate the paleo current flow direction.

Conclusion

A post stack seismic amplitude volume and its derivative products were successfully used for mapping glacial paleo valleys. Enhancing the signal of the traces is observed in the derived first and second derivative volumes. Furthermore, spectral decomposition analysis is applied to mix common frequencies in flattened stratigraphic slices. The final slices show changes in frequency interpreted to be geological change. The mapping workflow, divided into three phases: seismic time slices, cross sections, and the integration of non-seismic methods, provided a systematic approach to understanding the tunnel valley architecture. The flattening of the seismic cube on the Lower Pennsylvanian reflector (Upper Carboniferous) was a key step in restoring the depositional architecture, enabling a more accurate interpretation of the valley's geometry and infill. The use of four wells for calibration further validated the mapping process. This study demonstrates the effectiveness of high-resolution 3-D seismic data, combined with advanced seismic attributes and a structured mapping workflow, in characterizing tunnel valley systems. The integration of seismic and well data provided a comprehensive understanding of the tunnel valley's subsurface structure, offering valuable insights for regional studies. The methodology outlined in this study serves as a robust framework for similar investigations in other regions, emphasizing the importance of careful data conditioning and the integration of complementary datasets.

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